Lecture Notes on

The 2026 Nobel Prize in Physics

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These notes summarize the fourth Astrophysics Club meeting of the 2026-27 year. The 2026 Nobel Prize in Physics went to Francis Halzen for the IceCube Neutrino Observatory and the discovery of high-energy neutrinos from space. We start with what a neutrino is, then see how IceCube catches one, and finish with what IceCube found.

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1 The Prize

1.1 The Winner

- **Francis Halzen**, University of Wisconsin-Madison, born 1944 in Belgium.
- **Citation.** "For decisive contributions to the IceCube Neutrino Observatory and the discovery of high-energy neutrinos of astrophysical origin."
- **Fun fact.** It is the first solo physics prize since 1992.

1.2 The Question

- **Question.** What does this Nobel Prize even mean?
- **By the end.** The discovery changes observational astronomy forever.

2 Neutrinos

2.1 Through Your Fingernail

- **Question.** How many neutrinos from the Sun pass through your fingernail every second?
- **Answer.** 65 billion, or 65,000,000,000, every second.

2.2 The Ghost Particle

Neutrinos fly straight through the whole planet.

2.3 The Neutrino

The neutrino is written with the Greek letter ν.

- **Charge.** None.
- **Mass.** Almost none: less than a millionth of an electron's mass.
- **Flavours.** Three: the electron neutrino ν_e, the muon neutrino ν_μ, and the tau neutrino ν_τ.
- **Proposed** in 1930 by Wolfgang Pauli.
- **Detected** in 1956 by Cowan and Reines.

2.4 Four Fundamental Forces

- The four forces are gravity, electromagnetism, the strong force, and the weak force.
- Neutrinos only feel the weak nuclear force. They feel gravity too, but very weakly.

2.5 Why It Is a Ghost

Picture a blazar sending particles toward Earth.

- **Charged particles**, like electrons, are bent by magnetic fields through space, so they wander off course on the way.
- **A neutrino** goes in a straight line. Nothing bends or stops it.

2.6 It Points Back to Where It Came From

Neutrinos travel in completely straight lines, so they are easy to trace back to their source. Follow the line back from Earth and you land on the source, here the blazar.

3 IceCube

3.1 They Built a Detector to Catch Neutrinos

The detector is run from the IceCube Lab at the South Pole.

3.2 A Cubic Kilometre of Ice

- The detector fills 1 km^3 of ice.
- It starts 1,450 m below the surface and ends at 2,450 m, a 1 km tall block. Bedrock is at 2,820 m.
- 86 cables hang down into the ice with 60 light sensors on each cable, for $86 \times 60 = 5{,}160$ light sensors.

3.3 A Neutrino Hits the Ice

1. Almost all of them fly straight through.
2. Very rarely, one hits a nucleus.

3.4 The Weak Force Makes the Muon

- A muon neutrino ν_μ meets a neutron.
- They trade a **W boson**, the particle that carries the weak force.
- The neutrino comes out as a **muon** μ^-, and the neutron comes out as a proton.

In short, the neutrino trades a W boson with a neutron, generating a muon.

3.5 Out Comes a Muon

The muon shoots off through the ice and gives off blue light as it goes.

3.6 Faster Than Light in Ice

- Light in empty space travels at 300,000 km/s.
- The muon travels at almost 300,000 km/s.
- Light in ice only travels at 229,000 km/s.

So the muon outruns its own light.

3.7 A Cone of Blue Light

Because the muon outruns its light, the blue light trails behind it in a cone. As the cone sweeps through the ice, the sensors it reaches light up.

3.8 Cone Idea With Sound

The same cone idea shows up with sound. The slide shows an F/A-18 Hornet at nearly the speed of sound.

3.9 Reading the Light

- **Colour** shows when the light arrived: red came first, green came last.
- **A big ball** means lots of light.
- Red to green shows which way the muon went, which points back to the source.

3.10 Why the South Pole?

1. **Clear, dark ice.** Light travels far, and no sunlight reaches 1.5 km down.
2. **Free and huge.** A cubic kilometre of it, already there.
3. **The Earth is a shield.** Only neutrinos can come up through the planet.

4 What IceCube Found

4.1 2013: The First Cosmic Neutrinos

- Two events were named **Bert** and **Ernie**.
- **Bert.** 1.04 PeV $= 1.67 \times 10^{-4}$ J.
- **Ernie.** 1.14 PeV $= 1.83 \times 10^{-4}$ J.

They were the first sign of neutrinos from space.

4.2 2017: One Neutrino, One Blazar

On 22 September 2017:

1. IceCube caught one neutrino and sent out an alert.
2. Telescopes turned to that spot in the sky.
3. The Fermi telescope found a blazar flaring in gamma rays.

The blazar is **TXS 0506+056**, the first known source of cosmic neutrinos.

4.3 2023: The Milky Way in Neutrinos

IceCube made the first picture of our galaxy made with something other than light. On the slide, the neutrino map from IceCube sits on top of the Milky Way in visible light.

4.4 A New Way to See the Universe

- **Telescopes** see light, and light stops at the surface.
- **Neutrinos** come straight from the core.
- The slide shows the Sun photographed with neutrinos.
- This also works at night, because the neutrinos can travel through the Earth.

5 A 30-Year Bet

5.1 The Price Tag

- IceCube cost $279 million, mostly from the U.S. National Science Foundation.
- It was completed in December 2010.

5.2 The Timeline

- **1988.** Halzen and Learned propose a detector in Antarctic ice.
- **1993.** First sensors in the ice (AMANDA).
- **2010.** IceCube complete.
- **2013.** Cosmic neutrinos discovered.
- **2026.** Nobel Prize in Physics.

As Halzen put it:

> *"In retrospect, the most amazing thing is that we got funded for this project."*